

George Conradie

Senior Data Engineer

CONTACTS

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CORE SKILLS

Data Engineering & Architecture



Python & Modern Data Tools



Automation & Integration



SECTOR EXPERIENCE

- Financial Services
- Manufacturing
- E-Commerce
- Logistics
- Healthcare
- IoT / Telematics / Spatial Analytics

LANGUAGES

English ● ● ● ● ●

Afrikaans ● ● ● ● ●

DRIVERS LICENCE

Code EB

SUMMARY

I am a seasoned Senior Data Engineer with 20 years' experience across Microsoft data platforms (SQL Server, SSIS, SSRS) and deep expertise in ETL design, data warehousing, and performance optimization.

In the last couple of years I've been expanding into Python-based data engineering, automation, and applied AI/ML, combining robust structured-data foundations with modern intelligent-automation tools.

I am recognized for my pragmatic approach to problem-solving across finance, e-commerce, manufacturing, and IoT domains.

WORK EXPERIENCE

PBT Group — Senior SQL & ETL Engineer (2022 – Present)

- **Old Mutual:** Migrated legacy reports to Azure, applying data compression to cut storage cost & improve query speed.
- **Sanlam:** Resolved SQL concurrency and locking issues impacting production.
- **EuroChem:** Helped design and build enterprise-warehouse ETL framework; assisted with optimization and testing.
- Supported high-volume pipelines and contributed to C# + SSIS deployment automation.

FMI / Bidvest Life — BI Developer (2019 - 2022)

- Co-architected an Azure-ready data warehouse and standardized SSIS ETL templates from micro-services feeds.
- Consolidated policy, claims & commission data from 20+ systems with undocumented business logic.
- Automated JSON/API integrations (Mailchimp etc.) and introduced GitHub source control for BI assets.

Pointer SA — SQL Developer, R&D Team (2017 – 2022 and part-time)

- Modernized the company's telematics / IoT data platform for high-volume event ingestion and spatial analytics in SQL Server.
- Optimized processing of millions of GPS and sensor events daily from connected vehicles.
- Implemented spatial indexing for route tracking, proximity, and region-based queries.
- Built BI layers for real-time fleet dashboards and predictive maintenance analytics.

EARLIER CAREER (2000 – 2016)
EOH Microsoft Coastal — SQL Developer / Consultant (2011 – 2016)

- **Old Mutual, Mastercard PGS, and Red Cross:** Delivered enterprise data-warehouse and BI solutions (SSIS, SSRS).
- **Mastercard:** Designed multi-billion-row warehouse used for fraud-detection. Implemented a hash-based ETL comparison method to detect data changes efficiently — a technique later adopted more broadly.
- **Hospice:** PEPFAR MER compliant reporting.

Kalahari.net — Data Warehouse / SQL Developer (2004 – 2011)

- Developed ETL pipelines and data-quality systems for South Africa's largest online retailer of its era.
- Led migration from manual data loads to automated SSIS frameworks, improving integrity and load performance.

INDEPENDENT & RESEARCH PROJECTS (2018 – PRESENT)

- **VisionAssist** – Wearable AI navigation prototype (ESP32-S3 + Raspberry Pi 5 + ToF/IMU/Camera); runs offline captioning (BLIP) and speech (Piper TTS) via Python sensor fusion.
- **TrafficCount** – Two-stage YOLOv11 + LLM API vehicle-analytics pipeline deployed with ONNX Runtime + CUDA + OpenCV for real-time classification and line-crossing analysis.
- **MyAssistant** – Python + DuckDB agent with tiered memory, RAG, and WhatsApp integration (Playwright) using local LLMs for contextual automation.
- **Least-Visited Navigation Algorithm** – Python simulation using a custom heuristic achieving ~1.5× coverage efficiency per maze block; extendable to multi-agent and RL tests.

SKILLS SUMMARY

Data Engineering & Architecture

SQL Server • SSIS • T-SQL • Data Warehousing • ETL Framework Design • Query Optimization • Performance Tuning • Data Modeling • SSRS • Power BI

Python & Modern Data Tools

Python • Pandas • NumPy • DuckDB • PyODBC • FastAPI • Flask • Tkinter • OpenPyXL • Git/GitHub • Docker • Linux • VS Code

Automation & Integration

Process Automation • API Integration (REST/JSON) • Event-Driven Pipelines • Cloud Migration (Azure SQL, Synapse) • DevOps (Source Control, Deployment Automation) • C# SSIS Utilities

Applied AI / Machine Learning (Emerging Focus)

Feature Engineering • Model Integration (ONNX Runtime) • Computer Vision (OpenCV, YOLOv8, Faster R-CNN) • LLM/RAG Concepts • CUDA Optimization • Basic NLP & Text Analytics

IoT, Spatial & Edge Systems (Practical Exposure)

Telematics & Sensor Data Processing • Spatial Analytics (SQL Server Spatial) • ESP32 / Raspberry Pi Integration • Sensor Interfacing (ToF, IMU, Ultrasonic) • Low-Power Edge AI Concepts • Python Control & Simulation

EDUCATION & CERTIFICATIONS

QUALIFICATION	NQF	YEAR	INSTITUTION
BSc Information Technology & Robotics Focus - Machine Learning • Control Systems • AI Integration • Embedded Robotics	7	To complete in Q1 2026	Eduvos

REFERENCES

Available on request